

Search

- The binary search tree T is a decision tree, where the question asked at an internal node v is whether the search key k is less than, equal to, or greater than the key stored at v .

- Pseudocode:

Algorithm TreeSearch(k, v):

Input: A search key k and a node v of a binary search tree T .

Output: A node w of the subtree $T(v)$ of T rooted at v , such that either w is an internal node storing key k or w is the external node encountered in the inorder traversal of $T(v)$ after all the internal nodes with keys smaller than k and before all the internal nodes with keys greater than k .

if v is an external node **then**

return v

if $k = \text{key}(v)$ **then**

return v

else if $k < \text{key}(v)$ **then**

return TreeSearch($k, T.\text{leftChild}(v)$)

else

$\{ k > \text{key}(v) \}$

return TreeSearch($k, T.\text{rightChild}(v)$)

Search (cont.)

- A picture:

